

Regularization

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Math prerequisites for this lecture: You should know about:

- derivatives and optimization (Appendix C in Boyd and Vandenberghe)
- norm of a vector (Section I, Chapter 3 in Boyd and Vandenberghe)

Regularization

Penalty for model complexity

With no bounds on complexity of model, we can always get a model with zero training error on finite training set - overfitting.

Basic idea: apply penalty in loss function to discourage more complex models

This is an alternative way to manage the bias-variance tradeoff. The regularization “level” controls the “flexibility” of the model, and therefore, its position on the bias/variance curve.

Regularization vs. standard LS

Least squares estimate:

$$\hat{w} = \underset{w}{\operatorname{argmin}} MSE(w), \quad MSE(w) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Regularized estimate w / **regularizing function** $\phi(w)$:

$$\hat{w} = \underset{w}{\operatorname{argmin}} J(w), \quad J(w) = MSE(w) + \phi(w)$$

Common regularizers

Ridge regression (L2):

$$\phi(w) = \alpha \sum_{j=1}^d |w_j|^2$$

LASSO regression (L1):

$$\phi(w) = \alpha \sum_{j=1}^d |w_j|$$

Graphical representation

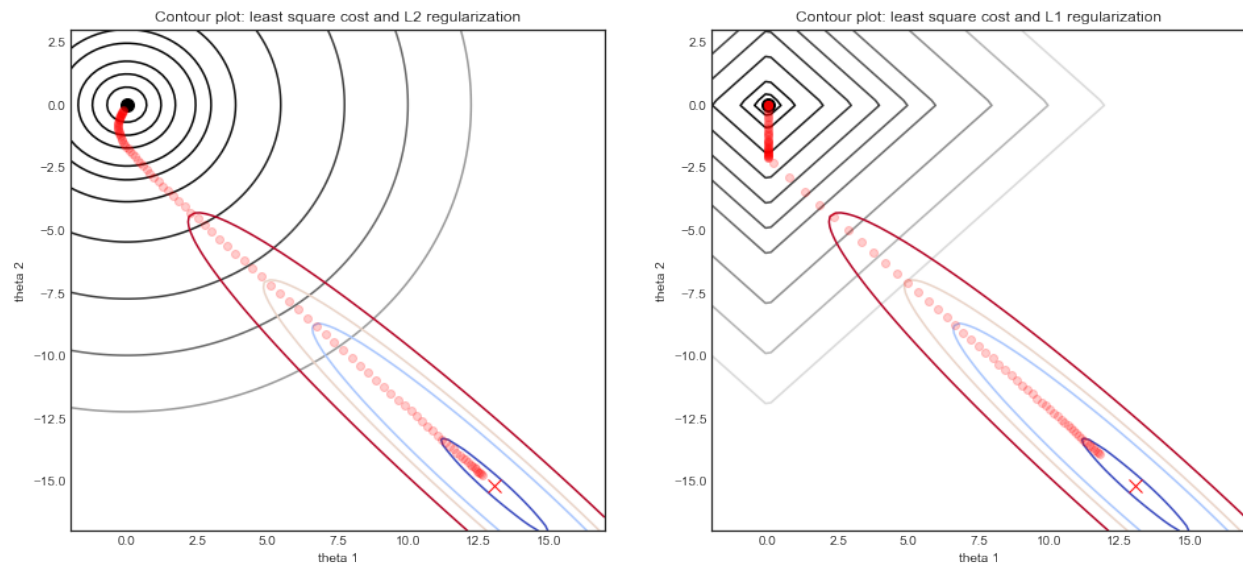


Figure 1: LS solution (+), RSS contours. As we increase α , regularized solution moves from LS to 0.

Common features: Ridge and LASSO

- Both penalize large w_j
- Both have hyperparameter α that controls level of regularization
- Intercept w_0 not included in regularization sum (starts at 1!), this depends on mean of y and should not be constrained.

Specific to Ridge (L2):

- minimizes $|w_j|^2$,
- minimal penalty for small non-zero coefficients
- heavily penalizes large coefficients
- tends to make many “small” coefficients
- Not for feature selection

Specific to LASSO (L1):

- minimizes $|w_j|$
- tends to make coefficients either 0 or large (sparse!)
- does feature selection (setting w_j to zero is equivalent to un-selecting feature)

To understand why L1 regularization tends to make sparse coefficients but not L2 regularization - look at the graphical representation. Note that the contours of the L1 regularization “stick out” when one or both parameters is zero.

Standardization

Before learning a model with regularization, we typically *standardize* each feature and target to have zero mean, unit variance:

- $x_{i,j} \rightarrow \frac{x_{i,j} - \bar{x}_j}{s_{x_j}}$
- $y_i \rightarrow \frac{y_i - \bar{y}}{s_y}$

Why?

- Without scaling, regularization depends on data range
- With mean removal, no longer need w_0 , so regularization term is just L1 or L2 norm of coefficient vector

Important note:

- Use mean, variance of *training data* to transform training data
- **Also** use mean, variance of *training data* to transform **test data**

Ridge with standardization

Assuming data standardized to zero mean, unit variance, the Ridge cost function is:

$$\begin{aligned} J(\mathbf{w}) &= \sum_{i=1}^N (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^d |w_j|^2 \\ &= \|\mathbf{A}\mathbf{w} - \mathbf{y}\|^2 + \alpha \|\mathbf{w}\|^2 \end{aligned}$$

LASSO with standardization

LASSO cost function ($\|\mathbf{w}\|_1$ is L1 norm):

$$\begin{aligned} J(\mathbf{w}) &= \sum_{i=1}^N (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^d |w_j| \\ &= \|\mathbf{A}\mathbf{w} - \mathbf{y}\|^2 + \alpha \|\mathbf{w}\|_1 \end{aligned}$$

Ridge term and derivative

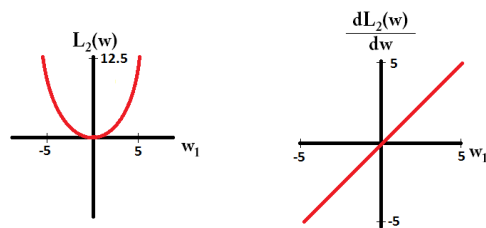


Figure 2: L2 term and its derivative for one parameter.

$$J(\mathbf{w}) = \|\mathbf{A}\mathbf{w} - \mathbf{y}\|^2 + \alpha\|\mathbf{w}\|^2$$

Taking derivative:

$$\frac{\partial J(\mathbf{w})}{\partial \mathbf{w}} = 2\mathbf{A}^T(\mathbf{y} - \mathbf{A}\mathbf{w}) + 2\alpha\mathbf{w}$$

Setting it to zero, we find

$$\mathbf{w}_{\text{ridge}} = (\mathbf{A}^T\mathbf{A} + \alpha\mathbf{I})^{-1}\mathbf{A}^T\mathbf{y}$$

LASSO term and derivative

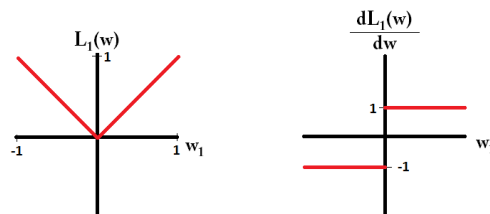


Figure 3: L1 term and its derivative for one parameter.

- No closed-form solution: derivative of $|w_j|$ is not continuous
- But there is a unique minimum, because cost function is convex, can solve iteratively

Effect of regularization level

Greater α , less complex model, more bias!

- Ridge: Greater α makes coefficients smaller.
- LASSO: Greater α makes more weights zero.

Selecting regularization level

How to select α ? by CV!

- Outer loop: loop over CV folds
- Inner loop: loop over α